Bishnu Prasad Kar

AI/ML Engineer – Machine Learning, LLMs, Agentic AI, Quantum Computing

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PROFILE

Computer Science undergraduate (B.Tech, expected August 2027) working across machine learning, large language models and emerging computation. Three conference papers presented (2024-2026). Builds systems end to end and reports results against baselines, with limitations stated plainly.

Research interests: Explainable and verifiable machine learning; LLM reasoning and multi-agent systems; retrieval-grounded generation; medical image segmentation; quantum machine learning (QAOA, quantum kernels); large-scale and streaming data systems.

EDUCATION

Qualification	Institution	Year
B.Tech, Computer Science and Engineering	Gandhi Institute of Excellent Technocrats, Bhubaneswar	Expected 2027
Higher Secondary (Class 12), Science	R.H.S.S, Mancheswar, Bhubaneswar	2020 - 2022
Secondary (Class 10)	S.S.V.M. Unit-3, Bhubaneswar	2020

CGPA 8.8/10 through semester 6 (currently semester 7). Coursework: Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Digital Logic Design.

CONFERENCE PAPERS

Presented at and certified by the conferences below; none published in a journal.

J. Dash, **B. P. Kar**, R. L. Mohanty

Entropy-Optimized Adaptive Hybrid Thresholding for Retinal Blood Vessel Segmentation

SPARK-2K26 Research Conclave, GIET Bhubaneswar, January 2026

Neural ODE-UNet manuscript: F1 0.797 and AUC 0.976 on DRIVE.

P. Mohanty, **B. P. Kar**, M. Jena, P. P. Upadhyay

CityFlow: A Scalable Big Data Framework for Real-Time Analytics in Computing 4.0

National Conference on Computing 4.0 (NCCENGT-2025), GIET Bhubaneswar, September 2025

Streaming ingestion, distributed storage and Spark processing, validated on the NYC Taxi dataset.

B. P. Kar, P. Panigrahi, S. N. Mishra

A Modular and Scalable Arithmetic and Logical Unit Design through Multiplexer-Based Controller

International Conference on Semiconductor Technologies and Renewable Energy (ICSTRE-2024), November 2024

8-bit ALU with multiplexer-based operation selection and timing verification.

INTERNSHIPS AND TRAINING

Data Science Intern Dec 2025 - Feb 2026

Jyesta Corporate Entity x E-Cell, IIT Roorkee

- Built recommendation and loan-prediction systems with feature engineering, several classifiers and explainability.

Trainee: Generative AI, Data Analytics, Cloud Computing 2024 - 2026

Central Tool Room and Training Centre (CTTC), Ministry of MSME

- Generative AI (2026): LLMs, Hugging Face, LangChain, RAG, end-to-end capstone. Data Analytics (2025): Python, SQL, Tableau.

Backend Engineering Trainee Jul 2025 - Jul 2026
Boot.dev (remote)

- Full backend track: Python, SQL, Linux, Docker, networking, system design.

MERN Stack Developer Intern Apr 2025 - May 2025
Codec Technologies India

- One-month project internship building full-stack web applications.

SELECTED PROJECTS

Q-CONSENSUS – Multi-Agent LLM Reasoning with Verifiable Decision Trails 2026 - present
Solo. Python, FastAPI, Qiskit Aer, llama.cpp, Solidity, Docker

- Five agents debate on a local LLM; a QAOA solver over quantum-kernel similarity picks the consensus answer, benchmarked against a classical baseline.
- SHA-256 hash-chained events anchored on a private Ethereum chain, so any run can be replayed and verified.

Misinformation Vaccine – Fake News Detection 2025
Solo. Python, PyTorch, Transformers, FastAPI, PostgreSQL

- DistilBERT fine-tuned on 43,944 articles: 99.95% accuracy, 0.9995 F1, against a 98.84% TF-IDF baseline.
- URL-to-verdict pipeline with stored history; dataset-bias limits documented rather than hidden.

Floodwatch – IoT Flood Early-Warning 2026
Team, Suravi 2K26 (1st position). Python, scikit-learn, SHAP, Ollama, Arduino

- 14 engineered features from three sensors feed a four-level risk model with SHAP attributions.
- A local LLM explains high-risk readings and the system triggers a hardware alert.

QuantumLab – Interactive Quantum Computing Lab 2026
Team Creative Cartel, Internal SIH, GIET 2026. Next.js, FastAPI, Qiskit Aer

- Circuits compile to Qiskit and run on Aer, so every probability and Bloch-sphere vector shown is simulated, never mocked. 15 routes, 9/9 correctness tests.

Project	Summary	Year
Soil-to-Shelf	Hyperledger Fabric traceability for Ayurvedic herbs, QR-based provenance (SIH, GIET)	2025
CycleSense	IoT and LLM menstrual health platform (3rd, ReFlow, IIT Bhubaneswar)	2025-26
ShelfSight	QR and OCR product scanning with a demand-prediction model	2025

TECHNICAL SKILLS

Programming, data	Python, SQL, Bash; pandas, NumPy, Matplotlib, Tableau, Advanced Excel
Machine learning	PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, NLP, computer vision, SHAP, model evaluation
Generative AI	LLMs, agentic systems, RAG, LangChain, FAISS, prompt design, llama.cpp, Ollama, Gemini API
Systems, platforms	FastAPI, Flask, REST APIs, PostgreSQL, MySQL, Hadoop, Spark, Docker, Git, Linux
Quantum computing	Qiskit, Qiskit Aer, QAOA, quantum kernels, variational circuits
Distributed ledger	Hyperledger Fabric, Solidity, Web3, SHA-256 hash chaining
Hardware, IoT	Arduino, ESP32, sensor integration, serial and MQTT pipelines
Languages	English, Hindi, Odia

AWARDS

- **1st position, Model Fiesta**, Suravi 2K26, GIET Bhubaneswar (Feb 2026) – Floodwatch flood early-warning system.
- **3rd place, ReFlow Menstrual Health Innovation Hackathon**, IIT Bhubaneswar (Nov 2025) – CycleSense.
- **Top performer**, 8-week Machine Learning training, Internshala – 94% final assessment (2025).

- **Completion with Merit**, one-month Quantum Computing course with workshops, C-DAC with IIT Roorkee – 81.67%.

CERTIFICATIONS

AI and deep learning	NVIDIA DLI, Getting Started with Deep Learning (2025); Harvard CS50 AI with Python (2025); IBM SkillsBuild (2025)
Generative AI	Microsoft and LinkedIn, Career Essentials (2024); Google Cloud (2025); CTTC, Ministry of MSME (2026)
ML and data science	Internshala, 8-week ML training, 94% (2025); University of Michigan, Data Science in Python (2024)
Big data	UC San Diego, four-course Big Data specialisation (2024-2025)
Quantum computing	C-DAC with IIT Roorkee, Quantum Computing with Merit, 81.67%
Simulations	Deloitte Data Analytics and KPMG Technology job simulations, Forage (2025)

HACKATHONS AND EVENTS

Hackathons	Internal Smart India Hackathon, GIET (2024, 2025, 2026, Team Creative Cartel); ReFlow, IIT Bhubaneswar (2025); Chakravyuh 1.0 national hackathon (2026); InnovateX 4.0 (2025)
Model Fiesta, Suravi	1st position 2K26; water quality prediction 2K25; participant 2K24
Events	AI Day Bhubaneswar 2025; AWS Summit India 2025; Google Gemini workshop 2025; machine learning workshops (PythonSoft, NSIC and MSME)